

A boundary-tested transcriptomic perturbation-triage framework identifies oligonucleotide validation candidates and fibrotic disease-state markers in idiopathic pulmonary fibrosis

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Abstract

Background: Idiopathic pulmonary fibrosis (IPF) is a progressive fibrotic lung disease with limited therapeutic options. Public transcriptomic datasets identify many reproducible IPF-associated signals, but only a subset are suitable for oligonucleotide-focused perturbation validation. We asked which reproducible IPF/fibrotic transcriptomic abnormalities should be advanced as perturbation-screening candidates and which should instead be triaged as disease-state markers, restoration hypotheses, or low-confidence regulatory hypotheses.

Results: Eight bulk mRNA or miRNA GEO datasets passed three-layer quality control, comprising 691 profiled samples. Cross-cohort screening retained 280 robust mRNA candidates and 10 robust miRNA candidates, defining a reproducible IPF/fibrotic transcriptomic evidence base rather than a stand-alone biomarker list. A leakage-

controlled Elastic Net score trained with discovery-only mRNA features showed strong external disease-state separation, but boundary stress tests using excluded GSE110147 NSIP and mixed IPF-NSIP samples, matched random-panel baselines, non-perfect-cohort summaries, and cohort-adjusted modeling indicated that the score captured a fibrotic interstitial-lung-disease state rather than IPF-specific diagnostic specificity. miRTarBase integration yielded 22 opposite-direction miRNA-mRNA candidate axes; strict evidence gates retained 3 exact mature hsa-miR-375 axes for main-text interpretation and downgraded 19 arm-agnostic axes to exploratory status. Target-program stress testing did not support expansion to a broader miRNA derepression program. Donor-aware single-cell pseudobulk validation localized key candidate signals, including SPP1 in myeloid cells, COL1A1/COL3A1/PTGFRN in stromal cells, CD24 in epithelial cells, and reduced GPX3 in stromal cells. The final perturbation-triage framework separated knockdown-screening candidates, context-dependent fibrotic disease-state markers, restoration/pathway markers, exact miRNA-axis hypotheses, and an externally motivated TNIK bridge.

Conclusions: This multi-cohort transcriptomic study provides a boundary-tested perturbation-triage framework for organizing IPF/fibrotic transcriptomic abnormalities before oligonucleotide-focused validation. The findings support epithelial, stromal, immune, and ciliary remodeling programs in IPF while emphasizing perturbation triage rather than diagnostic deployment or therapeutic-candidate validation.

Keywords

Idiopathic pulmonary fibrosis; oligonucleotide therapeutics; machine learning; miRNA; transcriptomics; single-cell RNA sequencing; candidate prioritization

Background

Idiopathic pulmonary fibrosis (IPF) is characterized by progressive distortion of lung architecture, aberrant epithelial repair, fibroblast activation, extracellular matrix accumulation, and immune remodeling [23,24]. Although antifibrotic therapies can slow functional decline, they do not reverse established disease, and there remains a need for mechanistically grounded molecular targets [23,24]. Oligonucleotide therapeutics provide a direct route to modulate transcripts or miRNA activity, but the translational value of a candidate depends on reproducibility across cohorts, compatibility with target directionality, and evidence that the candidate is active in relevant lung cell populations [27,28].

Public transcriptomic resources make it possible to evaluate these criteria systematically [1,2]. However, many IPF biomarker studies rely on single discovery cohorts, limited validation, or feature selection strategies that risk information leakage when validation data are used before model testing [29-34]. In addition, mRNA and miRNA evidence is often analyzed separately, making it difficult to nominate coordinated regulatory axes for oligonucleotide intervention [20,21].

Here, we designed a multi-cohort transcriptomic perturbation-triage study that combines GEO mRNA and miRNA datasets, external validation, miRTarBase target evidence, pathway enrichment, STRING protein interaction analysis, machine learning, disease-boundary stress testing, and single-cell localization [1,2,17-22,25,26]. The primary objective was not to propose another IPF biomarker classifier, but to determine which reproducible IPF/fibrotic transcriptomic signals are suitable for oligonucleotide-focused

perturbation validation and which should be downgraded as disease-state markers, restoration hypotheses, or low-confidence regulatory hypotheses.

Results

QC-controlled multi-cohort dataset and sample-label audit

The workflow integrated bulk mRNA, miRNA, and single-cell transcriptomic data from public GEO datasets (Figure 1) [1,2,25,26]. Eight bulk or miRNA datasets passed three independent quality-control checks: annotation completeness, expression-sample cross-matching, and matrix integrity. These datasets contained 495 bulk mRNA samples (304 IPF and 191 controls) and 196 miRNA samples (128 IPF and 68 controls). Two single-cell datasets were available for expression-level validation, while GSE122960 was excluded because a cell-level expression matrix was not available locally.

Table 1 summarizes the quality-controlled expression datasets.

series_id	data_type	dataset_role	samples	IPF/control	matrix_type	DE_method	QC
GSE110147	bulk mRNA	validation	33	22/11	microarray normalized log-intensity	limma	pass
GSE150910	bulk mRNA	validation	206	103/103	supplementary gene-level raw counts	edgeR-voom	pass
GSE21394	miRNA	validation	15	9/6	miRNA microarray normalized log-intensity	limma	pass
GSE27430	miRNA	validation	25	13/12	centered/transformed miRNA expression	limma	pass
GSE32537	bulk mRNA	discovery	169	119/50	microarray normalized log-intensity	limma	pass
GSE32538	miRNA	discovery	156	106/50	centered/transformed miRNA expression	limma	pass
GSE53845	bulk mRNA	validation	48	40/8	centered/transformed microarray expression	limma	pass

GSE92592	bulk mRNA	validation	39	20/19	supplementary mapped gene-count matrix	edgeR-voom	pass
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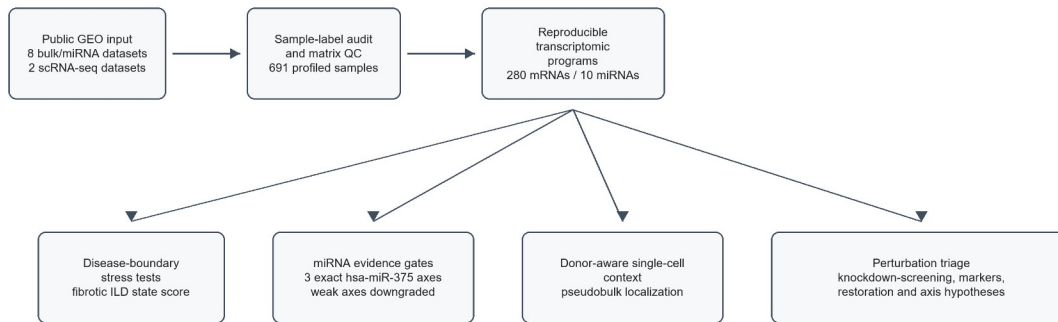
81

82 Dataset source publications were cited where available for the major bulk, miRNA, and
83 single-cell cohorts [6,25,26,36-41]. GEO accession records were retained in the
84 reference list to provide persistent dataset-level identifiers.

85 GSE110147 was treated as an IPF-vs-normal validation subset from a study that also
86 included NSIP and mixed IPF-NSIP samples. The main analysis included 22 IPF
87 samples and 11 normal controls; 10 NSIP and 5 mixed IPF-NSIP samples were
88 excluded before differential expression and machine-learning validation. Disease labels
89 were assigned from GEO sample metadata and manually checked against sample titles
90 and source descriptions; all included and excluded GSE110147 samples are listed in
91 the sample-level label audit in Additional file 1.

92 Sample-level inclusion, exclusion, and disease-label audit tables for all bulk and miRNA
93 datasets are provided in Additional file 1, including sample accession, original curated
94 disease label, final analysis label, inclusion status, exclusion reason, and source
95 annotation fields.

Boundary-tested perturbation-triage study design



All downstream gates use QC-passed datasets and preserve external validation cohorts for disease-state boundary testing.

Figure 1. Overview of the perturbation-triage study design. Public GEO mRNA, miRNA, and single-cell datasets were curated, audited, quality controlled, and analyzed through differential expression, cross-cohort robust candidate screening, disease-boundary testing, miRNA evidence grading, donor-aware single-cell pseudobulk validation, and perturbation-triage classification for oligonucleotide-focused validation planning.

Cross-cohort screening defines a reproducible IPF/fibrotic transcriptomic evidence base

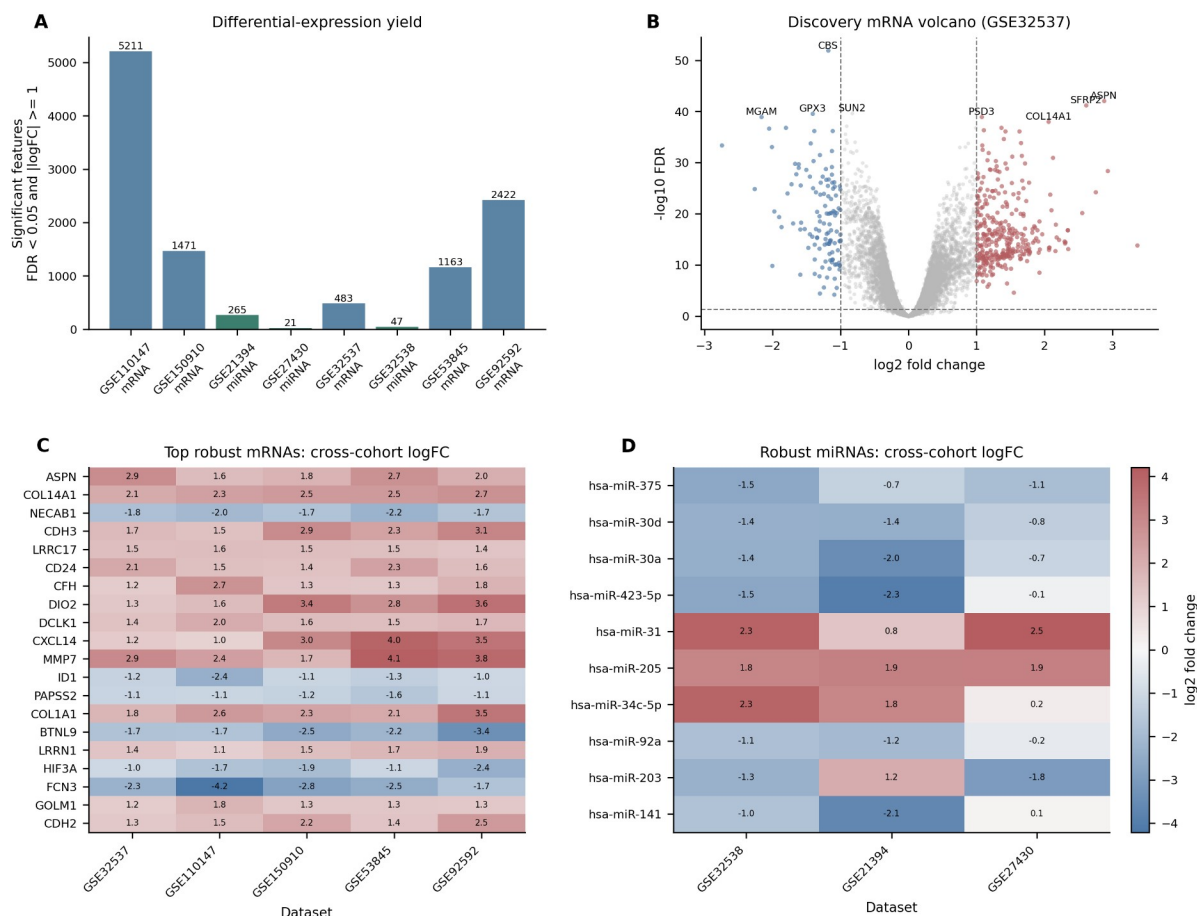
Differential expression was performed using limma for log-scale or normalized matrices and edgeR-limma voom for count-like matrices. In the mRNA discovery dataset GSE32537, 483 features passed $FDR < 0.05$ and absolute log fold change ≥ 1 ; after feature annotation and gene-level harmonization, 449 mRNA genes were used for discovery-validation screening. In the miRNA discovery dataset GSE32538, 47 significant miRNAs were detected. Direction-consistency screening across independent validation datasets retained 280 strict robust mRNA candidates and 10 strict robust miRNA candidates. The highest-

ranked robust mRNAs included ASPN, COL14A1, NECAB1, CDH3, LRRC17, CD24, CFH, DIO2, DCLK1, CXCL14, while the robust miRNA set included hsa-miR-375, hsa-miR-30d, hsa-miR-30a, hsa-miR-423-5p, hsa-miR-31, hsa-miR-205, hsa-miR-34c-5p, hsa-miR-92a, hsa-miR-203, hsa-miR-141. These candidates were treated as a reproducible IPF/fibrotic transcriptomic evidence base for downstream perturbation triage rather than as a final biomarker list. They were carried forward for candidate-axis construction, enrichment, network analysis, disease-state scoring, single-cell localization, and perturbation-triage classification.

Complete dataset-level differential-expression summaries, annotated feature-level results, and gene- or miRNA-level collapsed outputs are provided in Additional file 2. Supporting enrichment and protein-interaction analyses. GO, KEGG, and Reactome enrichment analyses used the gene-level feature universe tested in the GSE32537 discovery analysis as background [17,18]. For the 280 strict robust mRNAs, 251 symbols mapped successfully and 81 significant enrichment terms were detected. The most significant terms included cilium movement, cilium movement involved in cell motility, microtubule-based movement, and axoneme assembly. Terms containing sperm or flagellar labels were interpreted as shared axonemal/ciliary structural programs rather than reproductive biology.

STRING analysis mapped 267 of 280 robust mRNAs at medium confidence, yielding 950 edges and a largest connected component of 219 nodes [19]. The top-ranked hubs were DNAI1, TEK1, CFAP52, COL1A1, COL3A1, POSTN, RSPH4A, SPAG17, TTC25, COL6A3, COL1A2, VCAM1.

134 Complete GO, KEGG, Reactome, STRING mapping, edge, and hub metric outputs are
135 provided in Additional files 5 and 6.



136
137 Figure 2. Cross-cohort transcriptomic evidence base. (A) Number of significant differentially expressed
138 features in each QC-passed bulk mRNA or miRNA dataset. (B) Discovery mRNA volcano plot for
139 GSE32537. (C) Cross-cohort logFC/effect estimates for the top 20 robust mRNA candidates. (D) Cross-
140 cohort logFC/effect estimates for the 10 robust miRNA candidates. These robust signals were used as
141 the reproducible IPF/fibrotic transcriptomic evidence base for downstream triage rather than as a stand-
142 alone biomarker list.

143 **Disease-boundary testing supports a fibrotic ILD state score rather than**
144 **IPF-specific diagnostic specificity**

145 To avoid validation leakage, machine learning used only mRNA features selected from
146 the GSE32537 discovery analysis. After harmonizing gene-level matrices across the

147 discovery and external validation cohorts, 329 common discovery-only mRNA features
148 were available. Seven model families were evaluated with imputation, scaling, and
149 SelectKBest feature selection inside scikit-learn pipelines [22]; feature selection and
150 tuning were nested within resampling folds. Elastic Net achieved the highest mean
151 external ROC AUC. Across GSE110147, GSE150910, GSE53845, and GSE92592, the
152 best model had mean external ROC AUC 0.971, minimum external ROC AUC 0.910,
153 mean external PR AUC 0.964, and mean balanced accuracy 0.942. These headline
154 values are arithmetic means of per-cohort metrics unless otherwise stated. A label-
155 permutation control produced a mean AUC of 0.531, supporting non-random predictive
156 signal under the modeling workflow while not excluding cohort-composition or platform
157 effects.

158 The final stable panel contained 25 genes: ASPN, PTGFRN, COL14A1, GPX3, FNDC1,
159 NECAB1, CDH3, CFH, HMCN1, CD24, LRRC17, KLHL13, NPR1, CSF3R, SYTL2,
160 PSD3, SLCO4A1, PDE7B, GRASP, CBS, MGAM, IL1R2, GLT1D1, TNFRSF19,
161 TPPP3. Many top-ranked panel genes were also robustly direction-consistent across
162 validation datasets and were annotated with cautious follow-up labels, such as
163 conceptual siRNA or antisense knockdown compatibility for upregulated mRNAs. These
164 labels were used to organize validation priorities and should not be read as evidence of
165 therapeutic readiness. Pooled external interpretability analysis gave a ROC AUC of
166 0.924 across 326 validation samples, with GPX3 as the largest permutation-importance
167 feature followed by GRASP, CSF3R, CFH, ID1, FASN. Calibration and decision-curve
168 outputs were generated to support model-review discussions rather than to claim
169 immediate clinical deployability.

Two additional sensitivity checks addressed whether the high external performance was dominated by a single validation cohort or by generic robust-gene signal. Per-cohort Elastic Net ROC AUC values were 1.000 in GSE110147, 0.910 in GSE150910, 1.000 in GSE53845, and 0.974 in GSE92592. When each validation cohort was omitted from the summary, the mean cohort ROC AUC across the remaining cohorts ranged from 0.961 to 0.991 and the minimum retained cohort ROC AUC remained at least 0.910. In 500 random 25-gene panels sampled from the common robust mRNA universe and refit with the same fixed Elastic Net classifier, the mean external ROC AUC was 0.889 +/- 0.066, with a 95th percentile of 0.971. The observed 25-gene panel refit achieved mean external ROC AUC 0.964 and minimum external ROC AUC 0.907. These results support enrichment of predictive signal in the selected panel. At the same time, the random-panel analysis indicates that robust IPF-associated genes broadly carry disease-state information; therefore, the model is best interpreted as a cross-cohort disease-state prioritization tool rather than a clinically deployable classifier. Because the external ROC AUC values were high, we added disease-state boundary stress tests rather than additional model families (Additional file 8; Figure 3). Excluded GSE110147 NSIP and mixed IPF-NSIP samples were scored without using them for model training. Their scores were high and close to IPF samples (median score 0.930 for NSIP, 0.996 for mixed IPF-NSIP, 0.971 for IPF, and 0.095 for normal controls), indicating that the model captured a fibrotic interstitial-lung-disease state rather than IPF-specific diagnostic specificity. When validation cohorts with perfect ROC AUC were removed, the retained GSE150910 and GSE92592 cohorts still showed mean ROC AUC 0.942, mean PR AUC 0.929, and mean balanced accuracy 0.932. A cohort-

adjusted external logistic model retained an association between the Elastic Net logit score and disease status after validation-cohort fixed effects (coefficient 0.218; Wald P = 6.39e-24). In a matched random discovery-feature baseline, the observed final-panel refit had mean ROC AUC 0.946, close to the matched random-panel mean of 0.944 and below its 95th percentile of 0.968, while balanced accuracy remained at the high end of the matched distribution. These stress tests support use of the score as a disease-state prioritization layer, not as a uniquely optimized IPF diagnostic signature.

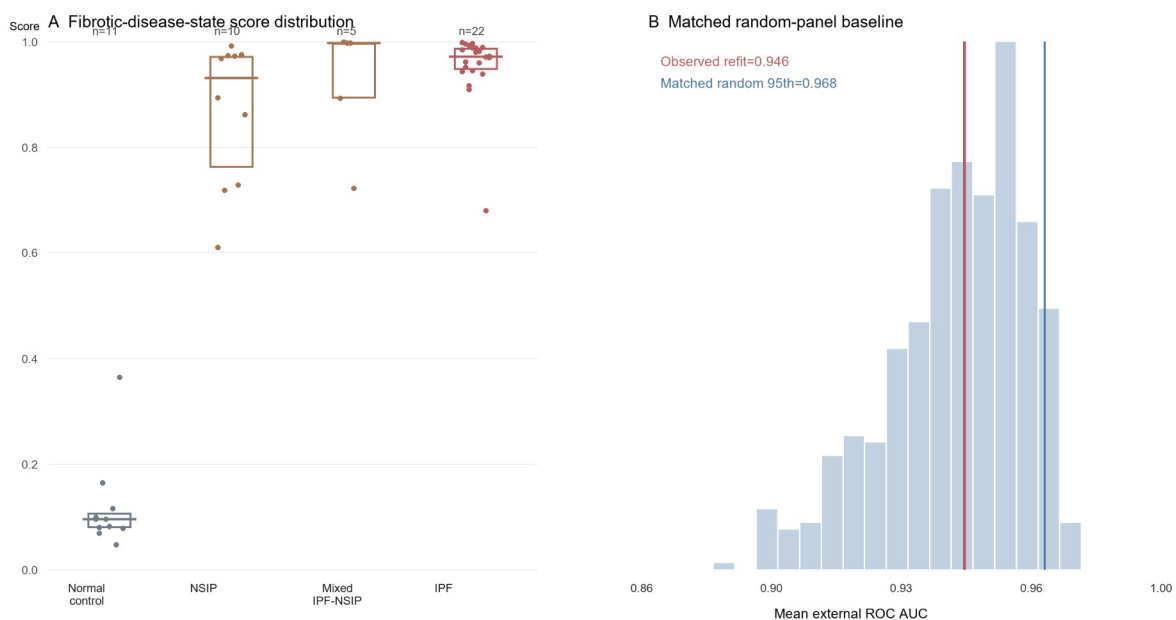
Table 3 summarizes the external performance and sensitivity checks. Per-cohort bootstrap 95% confidence intervals for ROC AUC and PR AUC, the full 25-gene panel, feature stability, predictions, hyperparameter grids, final Elastic Net specification, random-panel outputs, and disease-state boundary stress tests are provided in Additional files 7 and 8.

analysis	metric	value
Final Elastic Net	mean external ROC AUC	0.971
Final Elastic Net	minimum external ROC AUC	0.910
Final Elastic Net	mean external PR AUC	0.964
Final Elastic Net	mean balanced accuracy	0.942
Label permutation	mean ROC AUC	0.531
Leave-one-validation-cohort	retained mean ROC AUC range	0.961-0.991
Random robust 25-gene panels	mean external ROC AUC +/- SD	0.889 +/- 0.066
Observed 25-gene refit	mean external ROC AUC	0.964

Supporting reported-gene-set benchmarking. Because most published IPF machine-learning studies do not provide deployable model objects or full coefficients, we evaluated reported gene signatures as fixed external comparator feature sets [29-34]. Each comparator was trained only in GSE32537 and externally validated under the same framework used for the proposed model. The best published comparator, an

211 explainable machine-learning reported-gene set, achieved mean external ROC AUC
 212 0.946 [32]. Under this harmonized reported-gene-set benchmarking design, the
 213 proposed discovery-only Elastic Net panel showed higher mean external ROC AUC
 214 (0.971) and a higher minimum external ROC AUC (0.910), suggesting improved cross-
 215 cohort stability within this specific benchmarking framework. This comparison should be
 216 interpreted as reported gene-set benchmarking rather than direct comparison with fully
 217 deployable published models.
 218 Complete comparator metrics are provided in Additional file 7.

Figure 3. Disease-boundary stress tests for the fibrotic ILD state score



219 Figure 3. Disease-boundary stress tests for the fibrotic ILD state score. Excluded GSE110147 NSIP and
 220 mixed IPF-NSIP samples were scored without training use to evaluate whether the locked score behaved
 221 as an IPF-specific diagnostic separator or a broader fibrotic interstitial-lung-disease state score. Matched
 222 random discovery-feature panels contextualized the observed final-panel refit. These analyses define the
 223 interpretation boundary of the machine-learning layer rather than supporting diagnostic deployment.
 224

Strict miRNA evidence gates retain exact hsa-miR-375 axes and downgrade weak regulatory hypotheses

We integrated the robust miRNA and mRNA candidate sets with the human miRTarBase 2025 v10 interaction table [20,21]. Of 10 robust miRNAs and 280 robust mRNAs, 7936 miRTarBase records matched candidate miRNAs. Applying an opposite-direction rule between miRNA and mRNA discovery log fold changes identified 22 negative candidate axes involving 5 miRNAs and 16 target genes. Evidence grading separated 3 exact mature-miRNA matches from 19 arm-agnostic matches. hsa-miR-375 was downregulated in discovery (logFC -1.52) and showed same-direction decreases in both miRNA validation datasets (GSE21394 logFC -0.72; GSE27430 logFC -1.10). The exact hsa-miR-375 -> CLDN1, hsa-miR-375 -> MNS1, and hsa-miR-375 -> RPGRIP1L axes were retained as main-text prioritized hypotheses. The focused support table for these axes lists miRTarBase IDs, support type, validation method, PMID, miRNA/target species, and targeted cross-check status against accessible TargetScan/literature, miRDB, miRWalk, and CLIP-supported evidence. CLDN1 had additional literature support as a TargetScan-nominated and experimentally tested miR-375 target in lung cancer cells [35], whereas MNS1 and RPGRIP1L remained miRTarBase-derived exact-axis hypotheses without independently verified prediction or CLIP support in the current version. Arm-agnostic matches, including hsa-miR-92a with TP63 and hsa-miR-30a with CDH2, were retained only as exploratory supplementary hypotheses because they require mature-arm-specific validation before therapeutic interpretation. The exact mature-miRNA axes prioritized in the main text are shown in Table 2. Arm-agnostic axes are retained as exploratory hypotheses in Additional file 4.

axis	match_type	axis_score
hsa-miR-375 -> CLDN1	exact	49.000
hsa-miR-375 -> MNS1	exact	37.000
hsa-miR-375 -> RPGRIP1L	exact	37.000

248

249 Because exact mature-miRNA axes were intentionally stringent, we also tested whether
250 robust downregulated miRNAs showed broader target-program evidence among IPF-
251 upregulated robust mRNAs (Additional file 4; Figure 4). miRTarBase target sets for
252 robust downregulated miRNAs did not show FDR-supported enrichment in the IPF-
253 upregulated robust mRNA foreground. For hsa-miR-375, 250 miRTarBase exact or arm-
254 recoverable targets were present in the discovery mRNA background, but only 3
255 overlapped the upregulated robust mRNA foreground (CLDN1, MNS1, and RPGRIP1L;
256 odds ratio 0.63; FDR = 1.0). hsa-miR-375 targets also did not show a release-like
257 increase in discovery mRNA log fold change relative to non-target genes (one-sided
258 permutation P = 0.932). Therefore, the miRNA layer was not expanded beyond the
259 three exact hsa-miR-375 axes in the main interpretation; arm-agnostic and relaxed axes
260 remain exploratory.

Figure 4. miRNA evidence-gate stress tests

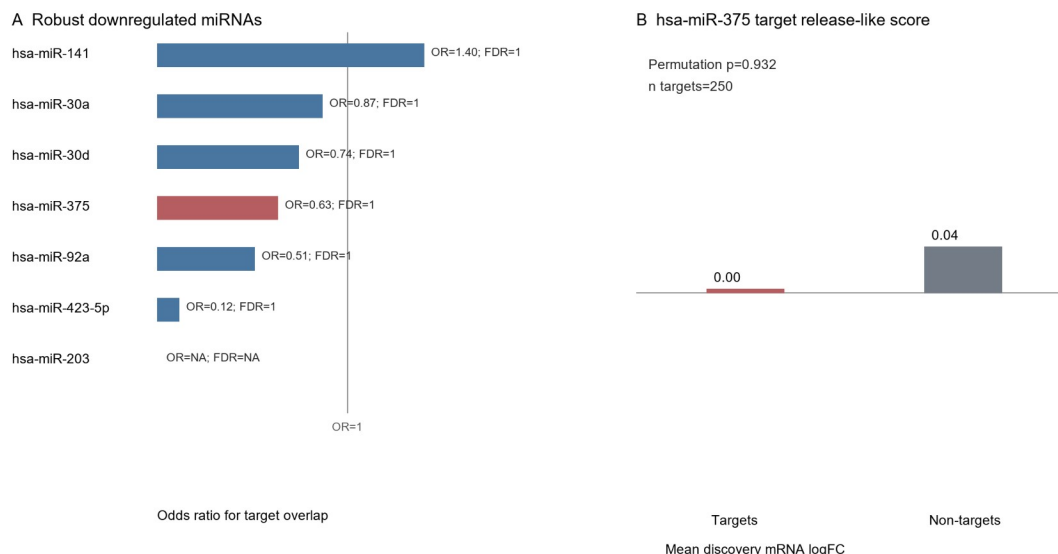


Figure 4. miRNA evidence-gate stress tests. Strict evidence grading retained exact mature hsa-miR-375 axes for main-text interpretation, while arm-agnostic axes were retained as exploratory hypotheses. miRTarBase target-set enrichment and the hsa-miR-375 target release-like score tested whether the exact-axis findings should be expanded to a broader miRNA target-program interpretation; the absence of FDR-supported target-program enrichment supported keeping the miRNA layer conservative.

Donor-aware single-cell pseudobulk localizes candidate signals

Single-cell validation was performed in GSE135893 and GSE136831, which together provided 332798 IPF/control metadata-matched cells [25,26]. The pipeline streamed sparse matrices directly, validated metadata alignment, and filtered non-informative cell labels including Multiplet, Outlier, MT-tRNAs, and CellCycle categories. Of 59 requested candidate genes, 59 were found in GSE135893 and 58 were found in GSE136831. The most prominent cell-type signals localized SPP1 to macrophage or myeloid compartments, COL1A1, COL1A2, COL3A1, and POSTN to stromal myofibroblast compartments, and GPX3 downregulation to fibroblast or myofibroblast populations. These cell-level summaries provide localization context rather than donor-level causal

277 inference. They support a disease model in which epithelial/ciliary changes, stromal
278 matrix remodeling, and macrophage activation converge in IPF and provide cellular
279 context for oligonucleotide-focused follow-up.

280 To reduce pseudoreplication concerns, we added a donor-aware pseudobulk sensitivity
281 analysis for core candidates (Figure 5C). Counts were aggregated at the donor/sample
282 x broad-celltype level before IPF-control comparison. Both single-cell datasets passed
283 donor metadata and matrix-integrity QC, with 22 donors in GSE135893 and 60 donors
284 in GSE136831. The pseudobulk analysis supported SPP1 upregulation in myeloid cells,
285 COL1A1/COL3A1 and PTGFRN upregulation in stromal cells, POSTN upregulation in
286 endothelial/stromal contexts, CD24 upregulation in epithelial cells, and GPX3
287 downregulation in stromal cells in GSE136831; COL14A1 was supported in
288 mesenchymal cells in GSE135893. POSTN pseudobulk support in the endothelial-
289 context comparison was interpreted as dataset-specific localization support, while
290 POSTN was retained overall as a context-dependent matrix/pathway candidate rather
291 than a simple knockdown-screening candidate. All displayed core pseudobulk
292 comparisons were direction-consistent with the bulk evidence and passed BH-adjusted
293 FDR < 0.01, with the largest donor-aware effects observed for SPP1 in myeloid cells,
294 POSTN in endothelial cells, COL1A1/COL3A1 in stromal cells, and reduced GPX3 in
295 stromal cells. These donor-level summaries strengthen the localization layer while
296 remaining validation-oriented rather than a full single-cell differential-expression study.
297 Complete single-cell localization summaries, donor-aware pseudobulk outputs, cell-
298 count summaries, and candidate-gene context tables are provided in Additional file 9.

Supporting immune-stromal module and single-cell communication analyses. To avoid treating the final candidates as isolated biomarkers, we added three exploratory mechanistic layers after the main discovery and validation analyses. First, coexpression modules were used to test whether robust and machine-learning candidates clustered into coordinated IPF-associated transcriptional programs. Second, curated ligand-receptor scoring was used to place prioritized genes in an intercellular signaling context. Third, a coexpression-neighborhood perturbation-priority proxy was used to organize knockdown-oriented follow-up candidates. These analyses were designed to support biological interpretation and experimental prioritization rather than to establish causal mechanisms.

The module analysis addressed whether the 25-gene panel and robust candidates were scattered classifier features or components of broader disease programs. In the GSE32537 discovery cohort, ten WGCNA-like coexpression modules were detected. M02 contained 211 robust mRNA candidates and 15 machine-learning panel genes, correlated with IPF status ($r = 0.66$), and aligned with epithelial/ciliary and matrix-remodeling marker scores. M09 contained 68 robust mRNA candidates, 10 machine-learning panel genes, and TNIK; it showed the strongest IPF correlation ($r = 0.74$). Thus, the module analysis provided a bridge between feature prioritization and pathway-level disease biology, while remaining an exploratory correlation analysis rather than a causal network model.

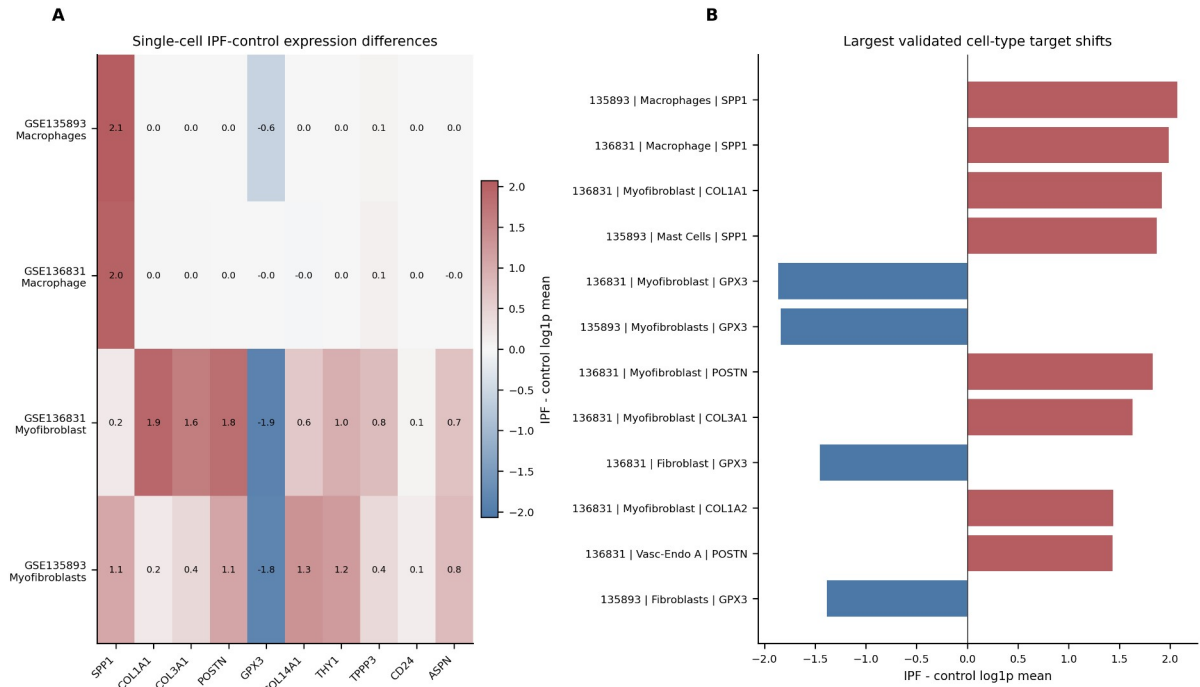
The curated ligand-receptor score was calculated as the product of mean normalized ligand expression in a sender cell type and mean normalized receptor expression in a receiver cell type, followed by the IPF-control difference for the corresponding sender-

322 receiver pair. A positive IPF-control score therefore indicates higher expression-based
323 communication potential in IPF than in controls, not a CellChat probability, statistical
324 interaction test, or experimentally validated signaling flux. After excluding non-
325 informative cell labels such as multiplets and outliers, 934 interaction changes were
326 scored. The strongest IPF-increased expression proxies involved MIF-CD74 across
327 endothelial, epithelial, and immune compartments; COL14A1-ITGB1 within
328 mesenchymal cells; CXCL12-CXCR4 from mesenchymal to immune cells; POSTN-
329 ITGB1; and SPP1-integrin/CD44 signaling. These results place the candidates in an
330 immune-stromal communication context but should be interpreted as hypothesis-
331 generating expression proxies.

332 Finally, the coexpression-neighborhood perturbation-priority proxy was not intended to
333 simulate molecular knockdown. Instead, it ranked whether suppressing an upregulated
334 candidate would be expected to align with reversal of its local IPF-associated
335 coexpression neighborhood in the discovery cohort. This proxy prioritized CD24,
336 COL14A1, POSTN, and PTGFRN as internally supported perturbation-screening
337 candidates, but POSTN was retained as a context-dependent matrix/pathway candidate
338 because it is a broad fibrotic matrix marker rather than a simple reversal target. SPP1
339 remained biologically important through macrophage/myeloid localization and ligand-
340 receptor evidence, but its negative proxy score indicates that it was not as well
341 supported as a simple bulk coexpression-neighborhood reversal target. TNIK had
342 validation-cohort differential-expression and Wnt/TNIK bridge evidence, but it was not
343 significant in the GSE32537 discovery analysis and was not selected by the final
344 machine-learning panel. Therefore, TNIK is best positioned as an externally supplied

345 translational hypothesis to test alongside the primary IPF target modules rather than as
346 a target discovered de novo by this study.

347 Complete coexpression module, curated ligand-receptor, coexpression-neighborhood
348 perturbation-priority, and TNIK evidence tables are provided in Additional file 10. The
349 main-text interpretation is intentionally limited to whether these outputs support the
350 primary candidate modules or motivate cautious external hypotheses.



C Donor-aware pseudobulk validation

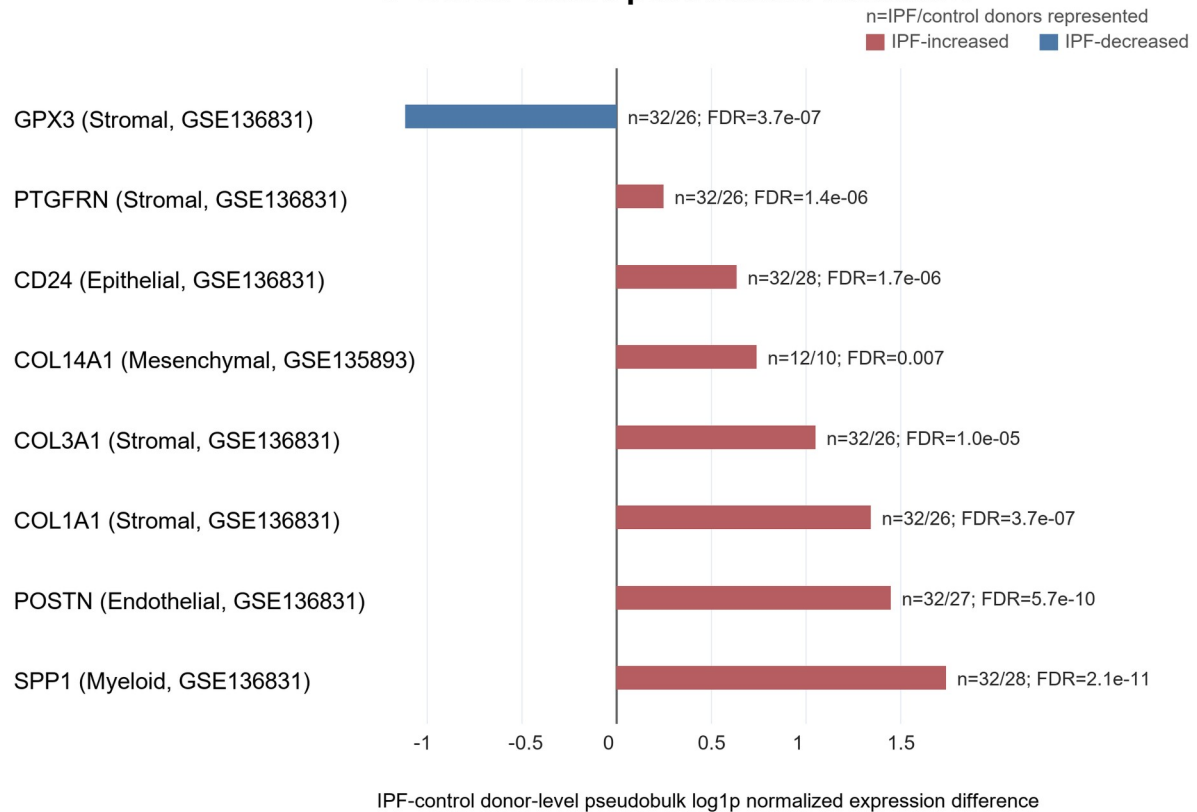


Figure 5. Donor-aware single-cell localization of candidate signals. (A) IPF-control expression differences for selected candidate genes across representative disease-relevant cell populations. (B) Largest cell-

type-level candidate shifts after filtering non-informative cell labels. (C) Donor-aware pseudobulk validation for core candidates. Red bars indicate IPF-increased pseudobulk expression and blue bars indicate IPF-decreased pseudobulk expression. The n labels denote the number of IPF/control donors represented in the indicated broad-celltype category after filtering, and FDR values are Benjamini-Hochberg adjusted across the tested core candidate x dataset x broad-celltype comparisons.

Perturbation-triage separates reversal-screening candidates from disease-state markers and restoration hypotheses

To convert the multi-omic results into a practical shortlist for oligonucleotide-focused follow-up, we built an evidence-weighted prioritization table that combined robust differential expression, machine-learning panel membership, PPI hub status, pathway membership, miRNA-axis evidence, single-cell localization, and direction-compatible oligonucleotide strategy labels. The priority score was designed to organize follow-up experiments, not to rank therapeutic promise directly. This distinction is important because upregulated candidates such as COL14A1, PTGFRN, CD24, ASPN, and CDH3 can be considered for knockdown screening, whereas broad matrix or inflammatory markers such as POSTN, COL1A1, COL3A1, and SPP1 require context-dependent pathway screening and downregulated candidates such as GPX3 and NECAB1 are better interpreted as pathway-state markers or restoration hypotheses. In total, 288 robust genes were scored, of which 10 were assigned Tier 1 priority. The highest-ranked genes were COL14A1, GPX3, ASPN, COL1A1, NECAB1, SPP1, PTGFRN, CD24, POSTN, CDH3.

To further separate this study from biomarker-only discovery, we converted the integrated evidence into an oligonucleotide-focused perturbation-triage map (Figure 6). This triage score did not add a new disease-association test; instead, it translated the

existing evidence layers into experimental validation classes. Knockdown-screening candidates included upregulated genes with cross-cohort support and a plausible knockdown-screening route, such as CD24, COL14A1, PTGFRN, ASPN, and CDH3. Context-dependent candidates, including POSTN, SPP1, COL1A1, and COL3A1, were retained as biologically important but not necessarily simple reversal candidates. GPX3 and NECAB1 were assigned to restoration or pathway-marker classes because they were downregulated in IPF. The exact hsa-miR-375 target genes CLDN1, MNS1, and RPGRIP1L were grouped as miRNA-axis hypotheses, and TNIK was retained as an external Wnt/TNIK bridge rather than a primary discovery candidate. The perturbation-triage score should be interpreted as a structured validation-planning heuristic, not a validated predictive model of oligonucleotide efficacy. Integrated priority score and perturbation-triage score serve different purposes. The integrated priority score ranks disease-association evidence across robust expression, machine-learning, PPI, enrichment, miRNA-axis, and single-cell layers. The perturbation-triage score reorganizes a subset of candidates by experimental route and direction-compatible validation logic. Therefore, the two rankings are not expected to be identical; for example, CD24 can rank highly by perturbation triage because it is upregulated, machine-learning-selected, cell-localized, and perturbation-priority supported, whereas COL14A1 ranks highly by integrated disease-association evidence. Because the perturbation-triage score uses manually specified validation-planning weights, we added a weight-sensitivity analysis. Candidate ranks were recalculated with equal weights, leave-one-evidence-layer-out scores, and 1000 seeded +/-20% perturbations of the base weights. The five knockdown-screening candidates remained

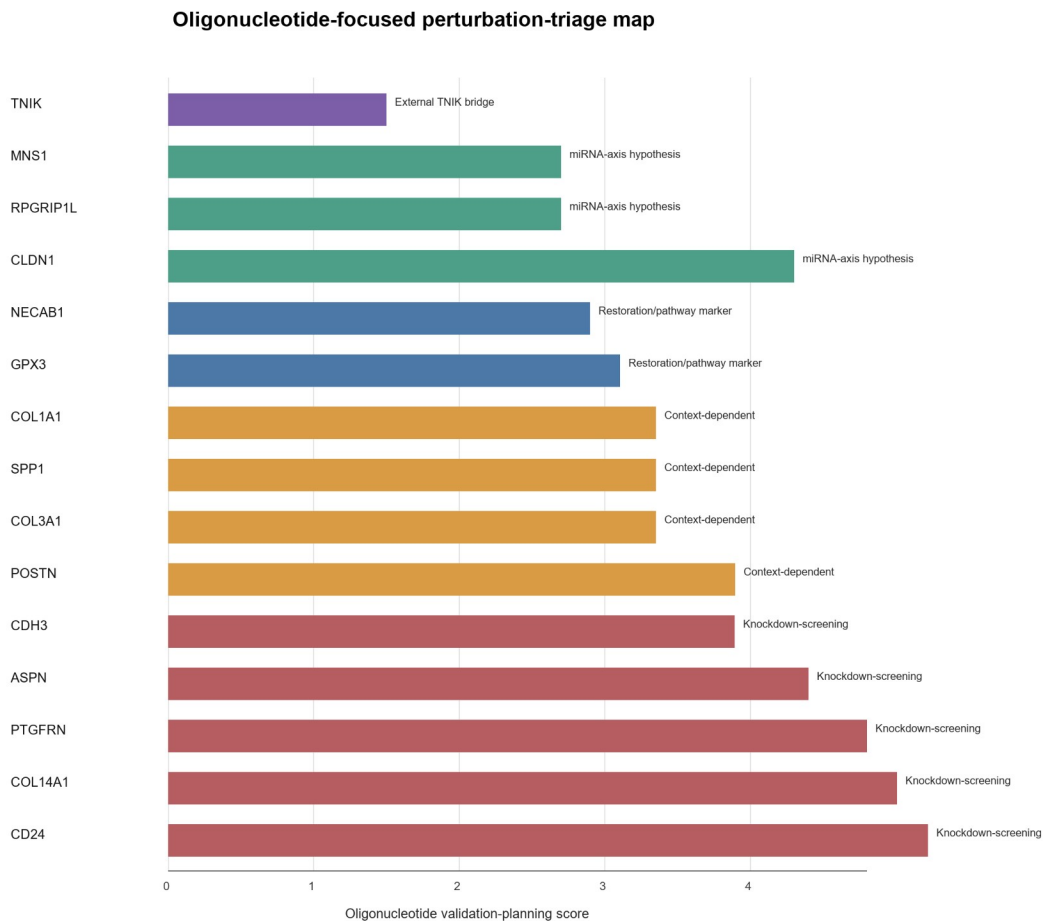
present across all sensitivity scenarios and each retained top-10 placement in at least 95% of scenarios, supporting the stability of the experimental class assignment while preserving the interpretation that the score is a prioritization aid rather than an efficacy estimate.

The final biological model organized the findings into five interpretable modules: macrophage/myeloid activation, stromal matrix remodeling, stromal antioxidant/metabolic loss, epithelial/ciliary remodeling, and an exact hsa-miR-375 target-axis hypothesis module. This organization separates biomarker classifier genes, disease-state markers, and putative perturbation candidates, which should not be treated as interchangeable categories.

Table 4 lists the top integrated candidate-prioritization results. Full scoring columns and additional candidates are provided in Additional file 7.

gene_symbol	direction	main evidence layers	cellular context	follow-up class
COL14A1	upregulated	robust; ML; PPI; single-cell	fibroblast/myofibroblast	knockdown screening
GPX3	downregulated	robust; ML; single-cell	fibroblast/myofibroblast	restoration or pathway marker
ASPN	upregulated	robust; ML; stromal module	pericyte/myofibroblast	knockdown screening
COL1A1	upregulated	robust; PPI; single-cell	myofibroblast/stromal	context-dependent matrix/pathway screening
NECAB1	downregulated	robust; ML	mesothelial/epithelial	pathway marker
SPP1	upregulated	robust; PPI; single-cell; LR	macrophage/myeloid	context-dependent inflammatory/pathway screening
PTGFRN	upregulated	robust; ML; module	myofibroblast	knockdown screening
CD24	upregulated	robust; ML; perturbation-priority proxy	AT2/transitional epithelial	knockdown screening
POSTN	upregulated	robust; PPI; LR; single-cell	myofibroblast/stromal	context-dependent matrix/pathway screening
CDH3	upregulated	robust; ML	goblet/epithelial	knockdown screening

414 The final biological model organized these findings into macrophage/myeloid activation,
415 stromal matrix remodeling, stromal antioxidant/metabolic loss, epithelial/ciliary
416 remodeling, and exact hsa-miR-375 target-axis hypothesis modules.



417
418 Figure 6. Perturbation-triage map. Candidate genes and miRNA-axis targets were organized into
419 knockdown-screening candidates, context-dependent fibrotic disease-state markers, restoration or
420 pathway markers, exact miRNA-axis hypotheses, and the externally motivated TNIK bridge. The
421 validation-planning score is intended to organize experiments and should not be interpreted as
422 therapeutic readiness or completed perturbation validation.

423 **Discussion**

424 This multi-cohort transcriptomic study integrates mRNA, miRNA, disease-boundary
425 testing, interaction-network, machine-learning, and single-cell evidence to triage

426 IPF/fibrotic transcriptomic abnormalities for oligonucleotide-focused follow-up. Several
427 points strengthen the resulting candidate set. First, mRNA and miRNA candidates were
428 not selected from a single cohort alone; instead, discovery signals were required to
429 show directionally consistent validation across independent datasets. Second, the
430 machine-learning workflow used discovery-only feature prefiltering and nested
431 resampling to reduce leakage, and the high-scoring output was explicitly bounded as a
432 fibrotic ILD disease-state score rather than a diagnostic deployment model. Third, strict
433 miRNA evidence gates and target-program stress tests prevented expansion beyond
434 exact mature-miRNA hypotheses when broader target-set support was absent. Finally,
435 donor-aware single-cell pseudobulk localized key candidates to biologically plausible
436 lung compartments.

437 The intended novelty of this work is not de novo discovery of individual IPF genes.
438 Several highly ranked candidates, including COL1A1, COL3A1, POSTN, SPP1, ASPN,
439 and COL14A1, are already biologically plausible in fibrotic lung disease. The distinction
440 from a biomarker-only analysis is the conversion of multi-cohort, direction-aware, cell-
441 contextual evidence into a perturbation-triage framework. This framing separates
442 knockdown-screening upregulated candidates, context-dependent matrix or
443 inflammatory disease-state markers, restoration/pathway-marker candidates, exact
444 miRNA-axis hypotheses, and the externally motivated TNIK bridge. The framework is
445 designed to organize validation experiments, not to imply that any candidate is already
446 therapeutically validated.

447 The supporting mechanistic extension adds a second layer of interpretation. Rather
448 than treating the disease-state panel as a black-box biomarker list, marker-score and

449 WGCNA-like module analysis showed that candidate genes concentrate in IPF-
450 associated epithelial/ciliary and matrix-remodeling modules. Curated ligand-receptor
451 scoring then connected these modules to altered intercellular signaling proxies,
452 particularly MIF-CD74, COL14A1-ITGB1, CXCL12-CXCR4, POSTN-ITGB1, and SPP1-
453 integrin/CD44 interactions. These results provide a coherent biological argument: the
454 nominated candidates are embedded in disease-associated epithelial, endothelial,
455 immune, and mesenchymal contexts. They remain hypothesis-generating and do not
456 establish causal cell-cell signaling.

457 The robust mRNA set captured several expected IPF-associated programs.

458 Upregulated matrix and fibroblast-related genes, including ASPN, COL14A1, COL1A1,
459 COL3A1, POSTN, and THY1, were supported by PPI and single-cell evidence. The
460 enrichment of cilium movement and axoneme assembly terms is also notable. Although
461 several ontology labels reference sperm motility, the shared genes encode axonemal
462 and microtubule-associated machinery relevant to motile cilia and epithelial biology.

463 This supports a broader interpretation involving epithelial remodeling and mucociliary
464 dysfunction rather than a reproductive process.

465 The miRNA analysis suggested several downregulated miRNAs with potential
466 replacement or modulation relevance. hsa-miR-375, hsa-miR-30a, hsa-miR-30d, and
467 hsa-miR-92a were directionally consistent across miRNA validation datasets, and
468 miRTarBase integration connected these miRNAs to upregulated mRNA candidates.

469 We explicitly separated exact mature-miRNA evidence from arm-agnostic evidence
470 because these categories do not carry the same evidentiary weight. The hsa-miR-375
471 -> CLDN1/MNS1/RPGRIP1L axes were exact mature-miRNA matches and therefore

472 suitable for main-text prioritization, whereas hsa-miR-30a and hsa-miR-92a axes remain
473 exploratory until mature-arm expression and target repression are validated in IPF-
474 relevant cells. The limited number of main-text miRNA axes therefore reflects strict
475 evidence grading rather than an attempt to claim a broad regulatory miRNA network.
476 These relationships should be considered prioritization hypotheses, because miRNA
477 effects are context-dependent and can involve many targets.

478 From an oligonucleotide-focused validation perspective, the final panel contains two
479 broad candidate classes. Upregulated mRNAs with stable validation and cell-type
480 localization may be considered for knockdown screening with siRNA or antisense
481 oligonucleotides. Downregulated miRNAs with inverse mRNA target relationships may
482 motivate miRNA mimic or pathway-restoration experiments. The integrated priority table
483 helps distinguish putative perturbation candidates such as COL14A1, ASPN, PTGFRN,
484 CD24, and CDH3 from pathway-state markers such as GPX3, where restoration or
485 pathway protection may be more biologically plausible than direct knockdown. However,
486 delivery to fibrotic lung compartments, target-cell uptake, off-target effects, and disease-
487 stage specificity remain major translational barriers.

488 The high external AUC values require cautious interpretation. Although validation
489 cohorts were not used for feature selection or model tuning, public transcriptomic
490 studies can still be influenced by platform, tissue-source, processing, clinical-annotation,
491 and cohort-composition differences. Boundary stress tests supported interpretation as a
492 fibrotic ILD disease-state score and a feature-prioritization layer, not as IPF-specific
493 diagnostic deployment. Although leave-one-validation-cohort summaries, disease-
494 boundary tests, and random robust-gene panel baselines were included, further

495 prospective validation and independent clinical-cohort testing would be needed before
496 clinical claims.

497 TNIK requires cautious interpretation. Public-data evidence placed TNIK in an IPF-
498 associated coexpression module and showed differential expression in several
499 validation cohorts, but it was not significant in the primary discovery cohort and was not
500 selected into the final machine-learning panel. Therefore, a TNIK-targeting small nucleic
501 acid can be connected to this manuscript only as an externally motivated translational
502 bridge, preferably through Wnt/TNIK pathway readouts and co-culture rescue
503 experiments. If the TNIK reagent is a knockdown oligonucleotide, the public IPF
504 transcriptomic direction does not by itself support a simple disease-reversal claim; its
505 value should be tested experimentally in the specific profibrotic cell state where TNIK
506 activity is hypothesized to be pathogenic.

507 Experimental follow-up should first confirm candidate expression and perturbation
508 efficiency in human lung fibroblasts, alveolar epithelial cells, and macrophage or
509 monocyte-derived macrophage co-culture models. Perturbation experiments under
510 TGF-beta1, epithelial-injury, or macrophage-conditioned contexts could then test
511 candidate siRNA/ASO or miRNA mimic effects on matrix deposition, epithelial injury,
512 inflammatory signaling, and Wnt/TNIK pathway readouts. These studies would be
513 required to move from the present prioritization map to experimental candidate
514 validation.

515 This study has limitations. It is retrospective and relies on public datasets generated
516 across different platforms, tissues, protocols, and clinical annotation schemes. Although
517 we applied strict sample matching and matrix QC, residual cohort effects cannot be

eliminated fully. The machine-learning model was externally validated across public cohorts but not prospectively validated, and very high external AUC values should be interpreted with attention to possible cohort structure. Published model comparison was based on reported gene signatures rather than original trained model objects, because most prior studies did not provide deployable model files or coefficients. The miRNA layer used miRTarBase evidence grading and a targeted cross-check table. CLDN1 had additional TargetScan/literature support outside miRTarBase, but MNS1 and RPGRIP1L lacked independently verified prediction or CLIP support in the current version; therefore, all exact mature-miRNA axes remain prioritization hypotheses rather than confirmed IPF regulatory mechanisms. Finally, single-cell validation supports cellular localization and donor-aware pseudobulk sensitivity for selected candidates, but it does not prove causal function. Experimental perturbation in relevant human lung cell systems and in vivo models is required before therapeutic interpretation.

Conclusions

The integrated analysis identified 280 robust mRNA candidates, 10 robust miRNA candidates, 22 opposite-direction miRNA-mRNA axes, and a 25-gene fibrotic disease-state panel with strong external separation but bounded diagnostic interpretation. Evidence grading retained three exact hsa-miR-375 target axes as miRNA-axis hypotheses and kept arm-agnostic axes as exploratory signals; target-program stress testing did not support expanding the miRNA interpretation beyond these strict axes. Perturbation triage highlighted validation classes that include knockdown-screening candidates such as COL14A1, ASPN, PTGFRN, CD24, and CDH3; context-dependent fibrotic disease-state markers such as COL1A1, SPP1, and POSTN;

restoration/pathway-marker candidates such as GPX3 and NECAB1; exact hsa-miR-375 axis hypotheses; and an external TNIK bridge. The evidence converges on ciliary/epithelial remodeling, extracellular matrix activation, macrophage-associated SPP1 signaling, and stromal GPX3 loss as important IPF-associated patterns. These results provide a boundary-tested perturbation-triage map for downstream oligonucleotide-focused validation.

Methods

Data collection and annotation

Public GEO datasets relevant to human IPF mRNA, miRNA, and single-cell transcriptomics were downloaded and organized locally. Non-human data were excluded. For each bulk or miRNA dataset, sample annotation files were curated with consistent identifiers, disease labels, data type, dataset role, and inclusion status. The expression matrix column identifier was treated as the primary sample_id, while GEO accession identifiers were retained separately when available.

Expression extraction and three-layer QC

Expression matrices were extracted from GEO series matrix files or supplementary count files [1,2]. Each dataset underwent three independent QC checks: annotation completeness, sample cross-match between annotation and expression columns, and expression-matrix integrity including numeric coercion, missingness, duplicate feature checks, and scale assessment. Only datasets passing all three checks were used in downstream bulk or miRNA analyses.

Matrix type was adjudicated from the GEO source, supplementary filename, expression range, zero fraction, and value distribution. GSE150910 and GSE92592 were treated as

564 supplementary gene-count matrices and analyzed with edgeR-limma voom.
565 GSE110147, GSE21394, and GSE32537 were treated as processed log-scale or
566 normalized intensity matrices and analyzed with limma. GSE27430, GSE32538, and
567 GSE53845 contained centered or transformed expression values, including negative
568 values; these datasets were analyzed with limma, and cross-cohort validation
569 emphasized FDR and direction consistency because absolute logFC values in centered
570 matrices are not directly interchangeable with raw count-derived log fold changes.
571 Dataset-level matrix-processing decisions are provided in Additional file 1.

572 **Differential expression**

573 Differential expression was performed in R using limma for processed log-scale or
574 normalized matrices and edgeR with limma-voom for supplementary count matrices [14-
575 16]. The primary contrast was IPF versus control. Discovery significance used FDR <
576 0.05 and absolute log fold change ≥ 1 . Probe IDs, transcript IDs, and miRNA probe
577 IDs were harmonized to gene symbols or miRNA names before cross-cohort
578 comparison. Mature-arm identifiers were retained where available; miRNA-family or
579 arm-unspecified records were flagged as arm-agnostic.

580 **Robust candidate selection**

581 GSE32537 served as the mRNA discovery cohort and GSE32538 as the miRNA
582 discovery cohort. mRNA validation used GSE110147, GSE150910, GSE53845, and
583 GSE92592. miRNA validation used GSE21394 and GSE27430. Strict robust mRNAs
584 required same-direction validation in at least two datasets. For log-scale or count-
585 derived datasets, validation support used FDR < 0.05 and absolute log fold change $\geq$
586 1; centered/transformed datasets were treated as direction-supporting validation layers

with cautious fold-change interpretation because their absolute logFC values are not directly interchangeable with raw count-derived log fold changes. Strict robust miRNAs required at least one same-direction validation dataset meeting the miRNA validation rule.

miRNA-mRNA target-axis integration

Robust miRNAs and mRNAs were integrated with the human miRTarBase 2025 v10 interaction table. Candidate axes were retained when the miRNA and mRNA discovery log fold changes were in opposite directions and the target gene was present in the robust mRNA set. Exact mature-miRNA matches and arm-agnostic matches were annotated separately.

Evidence grading and integrated candidate prioritization

miRNA-mRNA axes were graded by match specificity and experimental-support category. Exact mature-miRNA matches were eligible for main-text prioritization, while arm-agnostic matches were retained as exploratory hypotheses requiring mature-arm-specific validation. Robust mRNA candidates were then scored using a transparent additive framework incorporating robust-expression support, machine-learning panel evidence, PPI hub status, enrichment membership, miRNA-axis evidence, single-cell disease-control signal, and oligonucleotide strategy compatibility. The score was used to rank follow-up candidates, not to estimate causal effect size.

Perturbation-triage validation-planning score

The perturbation-triage validation-planning score was built from the integrated priority table, miRNA-axis evidence, single-cell localization summaries, and coexpression-neighborhood perturbation-priority proxy. It was defined as an additive validation-

610 planning score rather than a therapeutic-readiness score. The components were
611 disease direction (1.0 for upregulated candidates, 0.35 for downregulated candidates),
612 cross-cohort support (same-direction significant validation count divided by four and
613 capped at 1.0), single-cell localization (number of supporting single-cell datasets divided
614 by two and capped at 1.0), positive perturbation-priority support normalized to 0-1,
615 machine-learning selection frequency, and miRNA-axis support (exact mature-miRNA
616 evidence plus 0.25-weighted arm-agnostic evidence, capped at 1.0). Directionality and
617 cross-cohort support were assigned the highest weights because they determine
618 whether an oligonucleotide perturbation route is conceptually compatible and
619 reproducible across independent cohorts; cell localization and perturbation-priority
620 evidence were treated as intermediate validation layers, while machine-learning and
621 miRNA-axis evidence were used as supportive layers. The final score used weights of
622 1.30 for disease direction, 1.20 for cross-cohort support, 1.00 for cell localization, 1.00
623 for perturbation-priority support, 0.90 for machine-learning evidence, and 0.80 for
624 miRNA-axis evidence, with a 0.15 context penalty for broad matrix, inflammatory, or
625 downregulated markers where simple knockdown interpretation is less direct. In formula
626 form: $\text{score} = 1.30 \times \text{direction} + 1.20 \times \text{cross-cohort} + 1.00 \times \text{cell localization} + 1.00 \times$
627 $\text{perturbation-priority} + 0.90 \times \text{machine-learning} + 0.80 \times \text{miRNA-axis} - 0.15 \times \text{context}$
628 penalty . The 0.35 downregulated-candidate direction value and 0.15 context penalty
629 were chosen to encode validation-priority logic rather than learned from outcome data.
630 Candidate classes were assigned using the same evidence layers and biological
631 caveats. Upregulated candidates with robust validation and a plausible knockdown-
632 screening route were assigned to a knockdown-screening class. Broad matrix or

macrophage-associated genes with important disease biology but less straightforward perturbation interpretation were assigned to a context-dependent class. Downregulated genes were assigned to restoration or pathway-marker classes. Exact hsa-miR-375 target genes were assigned to the miRNA-axis hypothesis class, and TNIK was assigned to an external-bridge class rather than scored as a primary discovery candidate. The context penalty was binary and non-accumulating; a candidate received one 0.15 penalty if it fell into any broad matrix, inflammatory, or downregulated marker category with less direct simple-knockdown interpretation. The index was intended to organize experimental follow-up, not to infer therapeutic readiness or causal efficacy. Perturbation-triage score sensitivity was evaluated using three perturbation designs: equal-weight scoring, leave-one-evidence-layer-out scoring, and 1000 seeded random perturbations in which each base weight was multiplied by a factor sampled uniformly from 0.8 to 1.2. Candidate ranks and top-rank frequencies were summarized across all scenarios. This analysis tested whether the proposed validation classes were stable to modest changes in the manually specified score weights.

Marker-score modules, curated ligand-receptor scoring, and perturbation-priority proxy

Bulk immune, stromal, epithelial/ciliary, endothelial, and Wnt/TNIK marker scores were computed as within-dataset mean z scores of curated marker genes. In the GSE32537 discovery cohort, highly variable genes plus forced inclusion of robust candidates, machine-learning panel genes, ligand-receptor genes, and TNIK were clustered by correlation distance to generate WGCNA-like coexpression modules. Module eigengenes were calculated by singular-value decomposition and correlated with

disease status and marker scores. Single-cell communication potential was approximated by curated ligand-receptor scoring: mean normalized ligand expression in sender cell types was multiplied by mean normalized receptor expression in receiver cell types, and IPF-control score differences were calculated after excluding multiplet, outlier, cell-cycle, and mitochondrial tRNA labels. A coexpression-neighborhood perturbation-priority proxy ranked candidate perturbations by discovery-cohort coexpression-neighborhood structure and target directionality. These analyses are mechanism-generating extensions and should not be interpreted as outputs from full immune-deconvolution, formal cell-communication, or network-knockout packages.

Enrichment and PPI analysis

GO, KEGG, and Reactome enrichment analyses were performed with clusterProfiler and ReactomePA using the gene-level background tested in the GSE32537 discovery analysis. STRING protein interaction networks were queried for Homo sapiens at medium and high confidence thresholds. Hubs were ranked by combined degree, weighted degree, betweenness, and closeness metrics.

Machine learning and published-signature comparison

Gene-level matrices were generated by collapsing multiple probes or transcripts to gene symbols using within-dataset variance. The main machine-learning analysis used only discovery-significant mRNA features from GSE32537 that were common across external validation datasets. Seven model families were evaluated: lasso logistic regression, Elastic Net, linear support vector machine, radial-basis support vector machine, random forest, gradient boosting, and a small multilayer perceptron. Imputation used median replacement, scaling used StandardScaler, and feature

679 selection used SelectKBest with the f_{classif} score. Candidate feature counts were 15,
680 30, 50, and 100 where permitted by the available feature universe. Hyperparameter
681 tuning was performed inside GridSearchCV with ROC AUC scoring. Logistic and linear-
682 SVM C values were 0.03, 0.1, 0.3, and 1.0; Elastic Net additionally tested $l1_ratio$
683 values of 0.2, 0.5, and 0.8; RBF SVM tested C values of 0.1, 1.0, and 3.0 with gamma
684 set to scale; random forest tested max_depth values of 3, 5, and unrestricted with
685 $min_samples_leaf$ values of 1 and 3; gradient boosting tested $n_estimators$ of 80 and
686 150, learning rates of 0.03 and 0.08, and max_depth values of 1 and 2; the neural-
687 network classifier tested hidden-layer sizes of 16, 16/8, and 32/16 with alpha values of
688 0.001, 0.01, and 0.1. The internal design used repeated stratified five-fold outer cross-
689 validation repeated 10 times, with three-fold inner cross-validation for tuning. Final
690 models were refit in GSE32537 using five-fold internal tuning, and external validation
691 was performed only after model selection. Label permutation was used as a negative
692 control.

693 Published IPF signatures were evaluated as fixed gene-set comparators because
694 deployable trained model objects or full coefficients were generally unavailable [29-34].
695 Each comparator was trained in GSE32537 and externally validated on the same four
696 mRNA validation cohorts.

697 Model interpretability and utility checks were performed on pooled external validation
698 samples using permutation importance, calibration curves, and decision-curve analysis.
699 These outputs were used to assess whether model predictions were driven by
700 biologically plausible features and whether predicted probabilities had reviewable
701 calibration and threshold-utility behavior.

702 Sensitivity analyses used the locked Elastic Net external predictions for leave-one-
703 validation-cohort summaries. For the random robust-gene baseline, 500 random panels
704 with the same size as the observed 25-gene panel were sampled without replacement
705 from the common robust mRNA feature universe. Each random panel was trained in
706 GSE32537 using the final Elastic Net hyperparameters and externally evaluated in the
707 four validation cohorts. The observed 25-gene panel was refit with the same fixed
708 Elastic Net specification to provide an algorithm-matched comparison against the
709 random panels.

710 External disease-state boundary tests were added to define the interpretation of high
711 external AUC values. First, excluded NSIP and mixed IPF-NSIP samples from
712 GSE110147 were scored using a final-panel regularized refit without including these
713 samples in training or model selection. Second, 500 matched random discovery-feature
714 panels were generated by sampling 25 features from the common discovery-feature
715 universe while matching discovery direction, validation-support count, discovery
716 absolute logFC stratum, and discovery FDR stratum as closely as possible to the
717 observed panel. Third, external metrics were summarized after excluding validation
718 cohorts with perfect ROC AUC. Fourth, a pooled external logistic regression tested
719 disease status as a function of the Elastic Net logit score with validation cohort included
720 as a fixed effect. These analyses were intended to bound classifier interpretation rather
721 than to support clinical deployment.

722 **miRNA target-program stress tests**

723 Because exact mature-miRNA axis criteria were intentionally conservative, robust
724 downregulated miRNAs were also evaluated at the target-set level. Human miRTarBase

functional interactions were harmonized to exact or arm-recoverable mature-miRNA names and compared against IPF-upregulated robust mRNAs using the discovery mRNA background. For each robust downregulated miRNA, Fisher exact tests evaluated whether its miRTarBase target set was enriched among IPF-upregulated robust mRNAs, followed by Benjamini-Hochberg correction across tested miRNAs. For hsa-miR-375, a target release-like score compared discovery mRNA logFC values for hsa-miR-375 targets versus non-target genes and used 5000 seeded label permutations to estimate a one-sided enrichment P value. A relaxed exploratory axis set was also generated for supplementary audit by allowing exact or arm-recoverable miRNA matches with opposite-direction robust mRNA candidates. Paired miRNA-mRNA inverse correlation was not attempted because GSE32537 and GSE32538 did not provide explicit one-to-one shared donor identifiers in the available metadata.

Single-cell validation

Single-cell sparse expression matrices were processed by streaming matrix entries to avoid excessive memory use [25,26]. Candidate genes were evaluated by cell type and disease group. Cell labels corresponding to Multiplet, Outlier, MT-tRNAs, or CellCycle categories were excluded from manuscript-level summaries, and cell-type comparisons required at least 20 IPF and 20 control cells. These single-cell analyses were used as localization and prioritization evidence. They were not treated as causal differential-expression tests.

For donor-aware sensitivity analysis, core candidates were aggregated at the donor/sample x broad-celltype level in GSE135893 and GSE136831 after the same non-informative label filtering used for manuscript-level single-cell summaries.

748 GSE135893 donor IDs were taken from Sample_Name and GSE136831 donor IDs from
749 Subject_Identity. For each donor-celltype-gene combination, raw target-gene counts
750 were summed and normalized by the summed cell-level library size, then log1p
751 transformed. IPF-control differences were estimated across donors using two-sided
752 Welch tests within each dataset, broad cell type, and gene, with Benjamini-Hochberg
753 correction across all tested core candidate x dataset x broad-celltype pseudobulk
754 comparisons. This pseudobulk layer was restricted to core candidates and used as
755 validation support rather than a full single-cell differential-expression discovery analysis.
756 Figure 5C displayed direction-consistent comparisons with at least 10 IPF and 10
757 control donors and BH-adjusted FDR < 0.01. No additional per-donor minimum cell-
758 count threshold was imposed after broad-celltype aggregation; donor counts and per-
759 donor cell-count summaries are provided for audit. The n values shown in Figure 5C
760 denote the number of IPF/control donors represented in the indicated broad-celltype
761 category after filtering. The original-label to broad-celltype mapping and broad-celltype
762 cell-count summaries are provided in Additional file 9.

763 **Software and reproducibility**

764 Analyses were implemented with R, Python, Bioconductor packages including limma,
765 edgeR, clusterProfiler, ReactomePA, org.Hs.eg.db, and Python packages including
766 pandas, numpy, scikit-learn, scipy, matplotlib, and openpyxl. Software versions, random
767 seeds, script order, and input/output mapping are provided in Additional file 11. All
768 generated outputs are stored in the project results directory and are traceable through
769 the numbered scripts in the scripts directory.

770 **Declarations**

771 **Ethics approval and consent to participate**

772 Not applicable. This study analyzed publicly available de-identified datasets.

773 **Consent for publication**

774 Not applicable.

775 **Availability of data and materials**

776 All raw datasets analyzed in this study are publicly available from the Gene Expression
777 Omnibus under the accession numbers reported in Table 1 and the single-cell validation
778 section. Processed non-sensitive outputs supporting the conclusions are included as
779 Additional files 1-10 and Additional files 12-14. The numbered analysis scripts,
780 README file, software environment notes, and reproducibility instructions are provided
781 as Additional file 11 and are publicly available at
782 https://github.com/osbornzhou/IPF_oligo.

783 **Competing interests**

784 The authors declare that they have no competing interests.

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787 **Authors' contributions**

788 Yunyi Zhou curated the datasets, performed the computational analyses, generated the
789 figures and tables, and drafted the manuscript. Yanli Zhang supervised the study
790 design, interpretation, and manuscript revision. Both authors read and approved the
791 final manuscript.

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795 **Use of AI-assisted tools**

796 AI-assisted drafting and coding support were used to organize analysis scripts and
797 prepare an initial manuscript draft. The authors are responsible for verifying all
798 analyses, interpretations, citations, and the final submitted text.

799 **Additional files**

800 Additional file 1. XLSX. Dataset metadata and QC. Bulk, miRNA, and single-cell
801 annotation and expression-matrix QC tables.

802 Additional file 2. XLSX. Differential expression results. Annotated differential-expression
803 summaries and QC outputs.

804 Additional file 3. XLSX. Robust mRNA and miRNA candidates. Discovery-validation
805 robust candidate tables, direction matrices, and dataset-specific validation-support
806 rules.

807 Additional file 4. XLSX. miRNA-mRNA candidate axes. miRTarBase-derived opposite-
808 direction candidate axes, evidence grading, exact-axis support table, target-set stress
809 tests, hsa-miR-375 target release-like score, relaxed exploratory axes, and paired-
810 correlation audit.

811 Additional file 5. XLSX. GO/KEGG/Reactome enrichment results. Enrichment results,
812 gene-set inputs, and enrichment QC tables.

813 Additional file 6. XLSX. STRING PPI and hub results. STRING mapping, network
814 nodes, edges, hub metrics, and QC tables.

815 Additional file 7. XLSX. Machine-learning performance, feature stability, and
816 perturbation triage. Model performance, feature-selection stability, predictions,
817 hyperparameter grids, final Elastic Net specification, candidate-prioritization outputs,
818 perturbation-triage validation-planning score, and score weight-sensitivity analysis.

819 Additional file 8. XLSX. Machine-learning sensitivity analyses. Leave-one-validation-
820 cohort sensitivity, random robust-gene panel baseline outputs, disease-control score
821 stress test, matched random discovery-feature panel baseline, non-perfect-cohort
822 summary, and cohort-adjusted score test.

823 Additional file 9. XLSX. Single-cell localization summaries. Single-cell target-expression
824 summaries, donor-aware pseudobulk validation, broad-celltype mapping, localization
825 tables, and validation QC.

826 Additional file 10. XLSX. Mechanistic extension outputs. Marker scores, coexpression
827 modules, curated ligand-receptor scoring, coexpression-neighborhood perturbation-
828 priority proxy, and TNIK bridge tables.

829 Additional file 11. ZIP. Analysis code archive. Numbered analysis scripts, README file,
830 software environment notes, and reproducibility instructions.

831 Additional file 12. PDF. Donor-level pseudobulk plots. Supplementary donor-level
832 pseudobulk dot/median plots for the core single-cell validation comparisons shown in
833 Figure 5C.

834 Additional file 13. PDF. AUC disease-state stress-test plots. Supplementary disease-
835 control score distribution and matched random discovery-feature panel baseline for
836 external ML boundary assessment.

837 Additional file 14. PDF. miRNA target-program stress-test plots. Supplementary
838 miRTarBase target-set stress test and hsa-miR-375 target release-like score used to
839 bound miRNA interpretation.

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